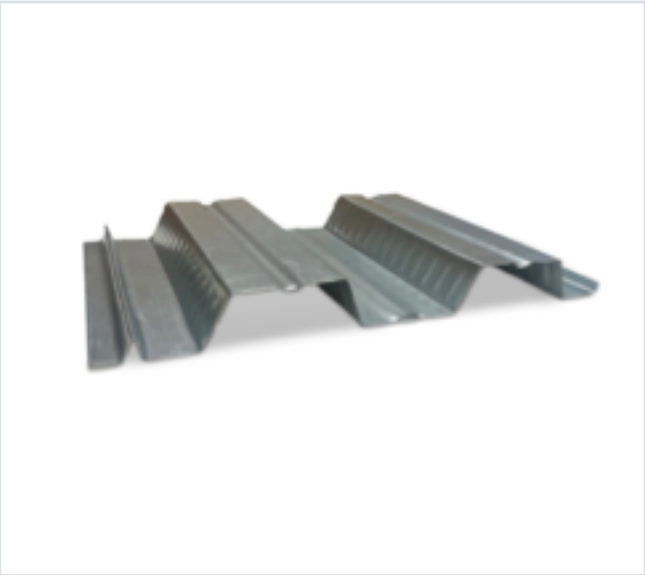


DECKING PRODUCT SUBMITTAL

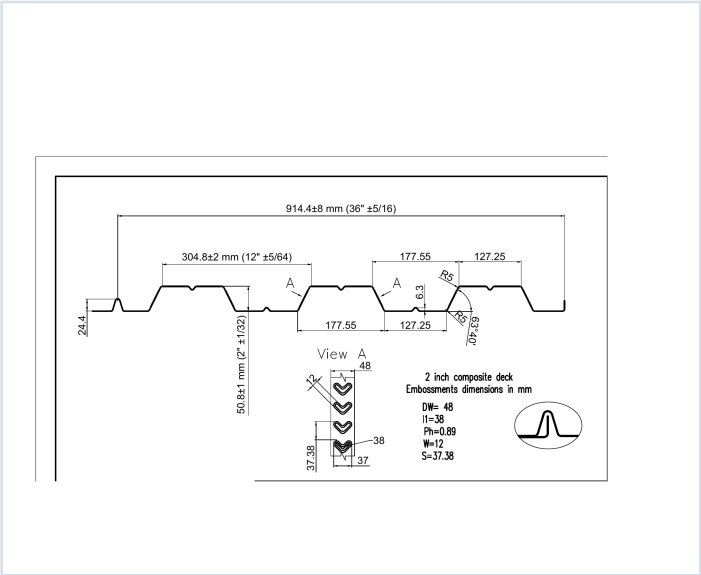
2CD-FY50-18-G90

PRODUCT	GRADE	GAUGE	COATING
2" Composite Deck	Grade 50	18 ga	G90

PRODUCT VIEW



PROFILE DRAWING



SECTION PROPERTIES · 18 GAUGE

Thickness (in.)	Weight (psf)	Fy (ksi)	S+ (in ³ /ft)	S- (in ³ /ft)	M+ (in-lb/ft)	M- (in-lb/ft)	I+ (in ⁴ /ft)	I- (in ⁴ /ft)
0.0474	2.5	50	0.493	0.500	14760	14960	0.548	0.530

SOURCE AND USE

1. Source: Refined Metal Steel Catalog 2025, published pages 54, 55, 56, 57, 58, 59, 60, 61
2. ASD section values above are the exact published row for this grade and gauge.
3. Coating selection does not change the published structural performance values.
4. Verify project-specific loading, fastening, span, concrete, and code requirements with the design professional.

PUBLISHED PERFORMANCE DATA

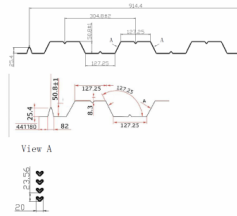
2CD-FY50-18-G90 · 18 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

Selected product row is reproduced on Page 1; excerpts retain published table context and notes.

STEEL CATALOG PAGE 54

GRADE 50 STEEL



Section Properties

Gage	Design Thickness (inches)	Weight (pcf)	E _x (ksi)	S _x (in ³)	S _y (in ³)	Se (in ⁴)	ASD (I _x + I _y) (in ⁴)	I _x + I _y (in ⁴)	I _x + I _y (in ⁴)
22	0.1095	16	50	0.247	0.254	2427	2427	0.31	0.290
20	0.0359	19	50	0.328	0.336	3860	3860	0.393	0.373
18	0.0474	25	50	0.485	0.500	14700	14700	0.548	0.530
16	0.0598	32	50	0.646	0.664	39100	39100	0.703	0.683

Note: All section properties and ASD flexural strengths are calculated in accordance with ANSI/AISC 360-10T, AISI S100-2012 and AISI S100-2016.

Shear and Web Crippling

Gage	V _u (kips)	Web Crippling (R _u /V _u) (ksi)		Web Crippling (R _u /V _u) (ksi)	
		End Bearing	Interior Bearing	End Bearing	Interior Bearing
22	388	434	476	529	627
20	278	295	325	364	424
18	365	386	424	484	574
16	463	507	557	634	754

Note: All section properties and ASD flexural strengths are calculated in accordance with ANSI/AISC 360-10T, AISI S100-2012 and AISI S100-2016.

STEEL CATALOG PAGE 55

Span	Gage	6'	7'	8'	9'	10'	11'	12'	13'	14'	15'	16'
Single	22	137	133	127	121	115	109	103	97	91	85	79
	20	183	174	165	156	147	138	129	120	111	102	93
	18	273	258	244	230	216	202	188	174	160	146	132
	16	368	345	323	301	279	257	235	213	191	169	147
Double	22	131	125	119	113	107	101	95	89	83	77	71
	20	186	177	168	159	150	141	132	123	114	105	96
	18	277	259	242	225	208	191	174	157	140	123	106
	16	367	342	318	294	270	246	222	198	174	150	126
Triple	22	176	169	162	155	148	141	134	127	120	113	106
	20	243	231	220	209	198	187	176	165	154	143	132
	18	346	324	303	282	261	240	219	198	177	156	135
	16	446	418	391	364	337	310	283	256	229	202	175

Notes:
1. All section properties and ASD (I_x + I_y) uniform loads are calculated in accordance with ANSI/AISC 360-10T, AISI S100-2012 and AISI S100-2016.
2. Loads shown in table are uniformly distributed superimposed loads in psf. Span lengths include end-to-end spacing of supports. Tabulated loads shall not be increased by increasing clear span dimensions.
3. Bending Moment formula used for flexural stress limitations are: Single and Two Span: $M_u \leq \frac{F_y S_x}{1.67}$; Three Span: $M_u \leq \frac{F_y S_x}{1.33}$.

Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

Uniform Superimposed Service Load that Causes L/240 Deflection (PSF)

Span	Gage	6'	7'	8'	9'	10'	11'	12'	13'	14'	15'	16'
Single	22	13	13	12	11	10	9	8	7	6	5	4
	20	13	13	12	11	10	9	8	7	6	5	4
	18	13	13	12	11	10	9	8	7	6	5	4
	16	13	13	12	11	10	9	8	7	6	5	4
Double	22	20	19	18	17	16	15	14	13	12	11	10
	20	20	19	18	17	16	15	14	13	12	11	10
	18	20	19	18	17	16	15	14	13	12	11	10
	16	20	19	18	17	16	15	14	13	12	11	10
Triple	22	24	23	22	21	20	19	18	17	16	15	14
	20	24	23	22	21	20	19	18	17	16	15	14
	18	24	23	22	21	20	19	18	17	16	15	14
	16	24	23	22	21	20	19	18	17	16	15	14

Note: For loads that cause L/120 Deflection, multiply by 2.0; For loads that cause L/180 Deflection, multiply by 1.5; For loads that cause L/360 Deflection, multiply by 0.667.

PUBLISHED PERFORMANCE DATA

2CD-FY50-18-G90 · 18 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

Selected product row is reproduced on Page 1; excerpts retain published table context and notes.

STEEL CATALOG PAGE 56

Normal Weight Concrete (145 pcf)				Lightweight Concrete (115 pcf)			
Maximum Unshored Clear Span				Maximum Unshored Clear Span			
400 (#200) 39 PSF	2x12x22 ga	7'-8"	9'-1"	9'-3"	2x12x22 ga	8'-6"	9'-9"
	2x12x20 ga	9'-3"	10'-6"	10'-10"	2x12x20 ga	10'-0"	11'-3"
	2x12x18 ga	10'-5"	12'-9"	12'-5"	2x12x18 ga	11'-2"	13'-9"
	2x12x16 ga	11'-4"	14'-2"	13'-4"	2x12x16 ga	12'-0"	14'-8"
450 (#250) 45 PSF	2x12x22 ga	7'-7"	8'-8"	8'-10"	2x12x22 ga	8'-3"	9'-5"
	2x12x20 ga	8'-10"	10'-0"	10'-4"	2x12x20 ga	9'-7"	10'-10"
	2x12x18 ga	9'-11"	12'-2"	12'-0"	2x12x18 ga	10'-9"	13'-3"
	2x12x16 ga	10'-11"	13'-8"	12'-10"	2x12x16 ga	11'-8"	14'-6"
500 (#300) 51 PSF	2x12x22 ga	7'-3"	8'-4"	8'-6"	2x12x22 ga	7'-8"	9'-1"
	2x12x20 ga	8'-6"	9'-7"	9'-11"	2x12x20 ga	9'-3"	10'-6"
	2x12x18 ga	9'-7"	11'-8"	11'-8"	2x12x18 ga	10'-5"	12'-9"
	2x12x16 ga	10'-6"	13'-3"	12'-6"	2x12x16 ga	11'-4"	14'-2"
550 (#350) 57 PSF	2x12x22 ga	6'-11"	8'-0"	8'-2"	2x12x22 ga	7'-7"	8'-9"
	2x12x20 ga	8'-3"	9'-3"	9'-6"	2x12x20 ga	8'-11"	10'-1"
	2x12x18 ga	9'-3"	11'-3"	11'-4"	2x12x18 ga	10'-0"	12'-4"
	2x12x16 ga	10'-1"	12'-9"	12'-2"	2x12x16 ga	11'-0"	13'-11"
600 (#400) 63 PSF	2x12x22 ga	6'-9"	7'-9"	7'-10"	2x12x22 ga	7'-5"	8'-6"
	2x12x20 ga	7'-8"	8'-11"	9'-2"	2x12x20 ga	8'-8"	9'-9"
	2x12x18 ga	8'-11"	10'-10"	11'-0"	2x12x18 ga	9'-9"	11'-10"
	2x12x16 ga	9'-9"	12'-4"	11'-10"	2x12x16 ga	10'-8"	13'-6"
650 (#450) 69 PSF	2x12x22 ga	6'-8"	7'-6"	7'-7"	2x12x22 ga	7'-2"	8'-3"
	2x12x20 ga	7'-9"	8'-7"	8'-10"	2x12x20 ga	8'-5"	9'-5"
	2x12x18 ga	8'-8"	10'-6"	10'-9"	2x12x18 ga	9'-5"	11'-6"
	2x12x16 ga	9'-6"	11'-8"	11'-7"	2x12x16 ga	10'-4"	12'-4"

Note
With crippling and clear span not shown accounted for in these tables. Required bearing should be determined based on specific span conditions.

STEEL CATALOG PAGE 61

Slab Thickness (Inches)	Weight (pcf)	7'-0"	7'-6"	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"
4	31	400	400	375	330	290	260	230
4.5	35	400	400	400	400	395	377	364
5	39	400	400	400	400	400	379	357
5.5	44	400	400	400	400	400	400	393
6	48	400	400	400	400	400	400	400
6.5	53	400	400	400	400	400	400	400

Slab Thickness (Inches)	10'-6"	11'-0"	11'-6"	12'-0"	12'-6"	13'-0"	13'-6"	14'-0"
4	290	280	272	257	243	230	220	210
4.5	296	290	285	269	256	243	232	220
5	304	295	290	272	259	246	235	223
5.5	314	305	300	281	268	255	244	232
6	324	315	310	291	278	265	254	242
6.5	334	325	320	301	288	275	264	252

16 ga Lightweight Concrete (115 pcf, F_c = 3,000 psi)

Slab Thickness (Inches)	Weight (pcf)	7'-0"	7'-6"	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"
4	31	400	400	400	400	392	373	350
4.5	35	400	400	400	400	400	380	357
5	39	400	400	400	400	400	400	377
5.5	44	400	400	400	400	400	400	400
6	48	400	400	400	400	400	400	400
6.5	53	400	400	400	400	400	400	400

Slab Thickness (Inches)	10'-6"	11'-0"	11'-6"	12'-0"	12'-6"	13'-0"	13'-6"	14'-0"
4	260	250	245	230	217	204	193	180
4.5	267	257	252	236	223	210	199	187
5	277	267	262	245	232	219	208	195
5.5	287	277	272	255	242	229	218	205
6	297	287	282	265	252	239	228	215
6.5	307	297	292	275	262	249	238	225